

DIHINI NIHINSA GALAPPATHTHI

Kurunegala | +94 – 702101128 | Email: Dihini.Nihinsa@gmail.com

LinkedIn: [Dihini.Nihinsa](https://www.linkedin.com/in/dihini-nihinsa) | GitHub: [DihiniNihinsa](https://github.com/DihiniNihinsa)

PROFESSIONAL PROFILE

Data Science undergraduate at SLIIT with a strong interest in data analytics and machine learning. Hands-on experience in data analysis, preprocessing, and basic model building through academic projects. Passionate about transforming data into meaningful insights to support decision-making and real-world problem solving.

PROJECTS

Water Tank Monitoring & Quality Assurance System

Technologies and Tools: Arduino Uno, IoT, Embedded Systems, C++, Sensors (Ultrasonic, pH, Raindrop)

Developed an Arduino-based IoT system for real-time monitoring of water levels, quality, and leakage to promote conservation.

- Integrated ultrasonic sensors for automated pump control and pH sensors for continuous water safety assessment.
- Implemented a leakage detection mechanism using moisture sensors and a centralized Arduino Uno unit to trigger critical status alerts.

Web-Based Medicine/Drugs Ordering System

Technologies and Tools: Java, Spring Boot, Spring Security, SQL, Thymeleaf, HTML, CSS, JavaScript

Engineered a full-stack application to streamline pharmaceutical transactions, including product catalogs and secure order processing.

- Developed a robust backend with role-based access control (RBAC) to manage distinct workflows for customers and admins.
- Applied the Data Transfer Object (DTO) pattern and layered architecture to optimize API communication and system reliability.

Vehicle Rental System

Technologies and Tools: Java Servlets, Apache Tomcat, MVC Architecture, OOP, HTML, CSS, JavaScript

Designed and developed a web-based management system to handle vehicle listings, rental requests, and automated bookings.

- Implemented Object-Oriented design principles to manage multiple vehicle types and complex rental workflows.
- Utilized MVC architecture and Java Servlets to ensure a clear separation of concerns and efficient CRUD operations.

Wine Quality Forecasting

Technologies and Tools: Python, scikit-learn, pandas, KNN, StandardScaler

Built a binary classification system predicting wine quality as "Good" or "Bad" based on physicochemical properties.

- Evaluated 6 ML models; Random Forest achieved the best accuracy at 81.9%, outperforming KNN and other baseline classifiers.

Pink Petals – AI Skincare Platform

Technologies and Tools: React+Vite, Spring Boot, Flask, XGBoost, SMOTE, MySQL

Developed an AI-powered skincare e-commerce platform with two integrated ML pipelines.

- Combined a 7-day sales forecast model (XGBoost, $R^2=0.78$) with a product recommendation classifier (ROC AUC=0.81) using SMOTE for class balance.

Data Analytics Internship Showcase

Technologies and Tools: Python, scikit-learn, NLTK, TextBlob, Power BI, Chart.js

Built an interactive single-page portfolio showcasing 7 end-to-end analytics projects from a Data Analytics Internship.

- Featured live Chart.js explorers across 6 datasets with 3-level project filtering for interactive data exploration.

JobTracker – AI Powered Job Application Tracker

Technologies and Tools: React.js, TailwindCSS, Node.js, Express, MongoDB, Python Flask, JWT

Built a smart job application management platform with AI resume analysis and real-time tracking.

- Combined an AI-driven resume analyzer with a live analytics dashboard and a dedicated admin portal for application oversight.

CinePass – Movie Ticket System

Technologies and Tools: MERN Stack, React Native, Node.js, Express.js, MongoDB, JWT

Developed a mobile-first cinema booking platform with real-time seat selection and admin management.

- Featured automated seat layout generation and promotional discount codes alongside full booking history tracking.

Highway Bus Tracker

Technologies and Tools: React Native, Expo Router, TypeScript, Safe Area Context, Reanimated

Built a full-stack mobile bus tracking app for Sri Lanka with live GPS tracking and a complete booking journey.

- Delivered route-aware live tracking and seat booking with payment flow across 10 screens using React Native + Expo.

EDUCATION

Sri Lanka Institute of Information Technology | 2024 – Present
BSc (Hons) in Information Technology – Data Science
Ku/ Maliyadeva Balika Vidyalyaya | 2009 – 2023
G.C.E. Advanced Level (2023/24) – Physical Science Stream – 2Cs 1C
G.C.E. Ordinary Level (2019) – 7As 1B 1C

CERTIFICATIONS

2026 AI Agents with MongoDB – *MongoDB Skill Badge*
2026 Building GenAI Apps Learning Badge Path – *MongoDB*
2026 MongoDB Data Modeling Path v1 – *MongoDB*
2025 AI/ML Engineer – Stage Certification 01
2025 AI/ML Engineer – Stage Certification 02
2024 Java Programming – *UniCAT*

SKILLS

Technical Skills
Programming – Python, Java, JavaScript, TypeScript, C/C++, Kotlin, SQL, HTML/CSS
Frameworks & Libraries – React, React Native, Node.js, Express, Spring Boot, Flask, TailwindCSS, Pandas, Scikit-learn, XGBoost, LangChain
Databases & Tools – MongoDB, MySQL, Git, JWT, Power BI, Chart.js
AI/ML – Machine Learning, Data Mining, NLP, LLM Integration, Information Retrieval, AI Agents (MongoDB)
Soft Skills
Problem Solving, Teamwork, Leadership, Effective Communication, Critical Thinking

EXTRA-CURRICULAR ACTIVITIES

- Batch Representative – SLIIT (2023 – Present)
- Cadet Platoon (CPL) – 2020
- Cadet Platoon (Cadet) – 2017 – 2019
- Member of School Chess Team (2009 – 2012)
- Member of Media Unit – 2014

CLUBS AND ACTIVITIES

- IEEE Student Branch of SLIIT
- Member of School Scrabble Team – 2014
- Member of School Swimming Team – 2014
- Little Friends – 2015
- Member of the School ISA (International School Award) Project – 2017
- Member of School Astronomical Society – 2020

REFERENCES

Ms. Uthpala Samarakoon
Senior Lecturer / Faculty of Computing
Department of Information Technology
Sri Lanka Institute of Information Technology
Phone: +94-11 754 4138
Email: uthpala.s@sliit.lk

Ms. R.A. Shalini Rupasinghe
Assistant Lecturer / Faculty of Computing
Department of Information Technology
Sri Lanka Institute of Information Technology
Phone: +94-11 754 4801
Email: shalini.r@sliit.lk